Week 3: The Cramer–Rao lower bound and other properties of estimators

F79MA Statistical Models A

Heriot-Watt University

The Cramer–Rao Lower Bound

As we improve our understanding of statistical estimators, two natural questions arise: *for a given estimation problem, is there a fundamental limit to estimation accuracy that no estimator can overcome? If so, which estimators attain this limit and hence cannot be outperformed?*

We study these questions from the perspective of the estimation mean square error, which can be analysed in terms of estimation bias and variance (recall that, for any estimator T, $\mathrm{MSE}(T) = \mathrm{Var}(T) + B(T)^2$ from the linearity of the expectation operator).

The Cramer–Rao Lower Bound (CRLB) is a theoretical result stating that there is a fundamental limit (a lower bound) for the estimation variance. This bound stems from the finite amount of information that the data has about the parameter of interest - it is not related to a specific estimation method. We present this result for the case of estimators that are unbiased and smooth (continuously differentiable), and by assuming some mild regularity assumptions on the statistical model that allow us to simplify the proof. This result can be easily generalised to biased estimators *(see later)*. Generalisations to non-smooth models and estimators are highly technical and not covered in this course.

Before we state the CRLB, let $\underline{X}$ be a random sample from a distribution with density function $f_X(x;\theta)$ and let us consider the following two regularity assumptions:

Regularity assumption A1: The density function $f_{\underline{X}}(\underline{x};\theta)$ and its partial derivative $\frac{\partial}{\partial\theta} f_{\underline{X}}(\underline{x};\theta)$ are continuous in $\underline{x}$ and θ.

Remark 1: under A1, we can apply the Leibniz integral rule as follows

$$\int \frac{\partial}{\partial\theta} f_{\underline{X}}(\underline{x};\theta)\, d\underline{x} = \frac{\partial}{\partial\theta} \int f_{\underline{X}}(\underline{x};\theta)\, d\underline{x}\,, \tag{1}$$

(i.e., to interchange the differentiation and integration).

Regularity assumption A2: The function $\frac{\partial}{\partial\theta} f_X(\underline{x};\theta)$ and its partial derivative $\frac{\partial^2}{\partial\theta^2} f_{\underline{X}}(\underline{x};\theta)$ are continuous in $\underline{x}$ and θ.

The information $I(\theta)$ For a random sample $\underline{X}$ from a distribution with density function $f_X(x;\theta)$, we define the amount of information about θ, if it exists, as follows:

$$I(\theta) \triangleq \mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{X};\theta)\right]^2\right).$$

Under regularity assumptions A1 and A2, we can use the equivalent expression

$$I(\theta) = -\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{\underline{X}}(\underline{X};\theta)\right).$$

Remark 2: the proof of why this alternative formulation is true under A1 and A2 is given in the appendix.

The Cramer–Rao Lower Bound: Let $\underline{X}$ be a random sample from a distribution with density function $f_X(x;\theta)$. Let $T(\underline{X})$ be an unbiased estimator of θ. Assuming $I(\theta)$ exists, the regularity condition A1 on $f_X(x;\theta)$, and a regularity condition regarding $f_X(x;\theta)$ and $T(\underline{X})$ (detail in the proof), we have the bound

$$\mathbf{Var}\ (T(\underline{X})) \geq \frac{1}{I(\theta)} = \frac{1}{\mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{X};\theta)\right]^2\right)}.$$

In addition, under A2, the bound can be expressed as

$$\mathbf{Var}\ (T(\underline{X})) \geq \frac{-1}{\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{\underline{X}}(\underline{X};\theta)\right)}.$$

This form of the CRLB is often easier to calculate.

Remark 3: Under suitable regularity conditions, if the elements of the sample $\underline{X} = X_1,\ldots,X_n$ *are independent given* θ*, then* $f_{\underline{X}}(\underline{X};\theta) = \prod_{i=1}^n f_{X_i}(X_i;\theta)$ *and therefore*

$$\boldsymbol{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{X};\theta)\right]^2\right) = -\sum_{i=1}^n \boldsymbol{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{X_i}(X_i;\theta)\right).$$

If, in addition, the elements of the sample are identically distributed, then

$$\boldsymbol{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{X};\theta)\right]^2\right) = -n\boldsymbol{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{X_1}(X_1;\theta)\right),$$

further simplifying the calculation of $I(\theta)$ *and the CRLB for unbiased estimators of* θ*.*

Example 3.1 Let $\underline{X}$ be a random sample from the $\mathrm{N}(\mu, \sigma^2)$ distribution where σ^2 is known. We will determine the CRLB for μ and comment on $\overline{X}$ as an estimator of μ. We will assume that all the required regularity conditions are satisfied.

We have:

$$L(\mu) = L(\mu; \underline{X}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum(X_i - \mu)^2\right)$$

$$\Rightarrow l(\mu; \underline{X}) = \text{constant} - \frac{1}{2\sigma^2}\sum(X_i - \mu)^2$$

$$\Rightarrow U(\mu, \underline{X}) = \frac{1}{\sigma^2}\sum(X_i - \mu) = \frac{n}{\sigma^2}(\overline{X} - \mu).$$

So

$$\mathbf{E}(U^2) = \left(\frac{n}{\sigma^2}\right)^2 \mathbf{Var}\,(\overline{X}) = \frac{n}{\sigma^2}.$$

Then the CRLB is $\mathbf{Var}\,(T) \geq \frac{\sigma^2}{n}$.

Alternatively, note that

$$\frac{\partial^2}{\partial\mu^2} l(\mu; \underline{X}) = -\frac{n}{\sigma^2}.$$

Remark: $\overline{X}$ is unbiased and $\mathbf{Var}\,(\overline{X}) = \sigma^2/n$, i.e., $\overline{X}$ attains the CRLB. Therefore, within the class of smooth unbiased estimators for μ, $\overline{X}$ is optimal in terms of MSE performance. □

Example 3.2 Let $\underline{X}$ be a random sample from $\mathrm{N}(\mu, \sigma^2)$ where μ is known. We will determine the CRLB for σ^2 and comment on S^2 as an estimator of σ^2. Again, assume that all the required regularity conditions are satisfied.

We have

$$L(\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum(X_i - \mu)^2\right)$$

$$\Rightarrow l(\sigma^2) = \text{constant} - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(X_i - \mu)^2$$

$$\Rightarrow U(\sigma^2, \underline{X}) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(X_i - \mu)^2.$$

The approach using the second derivative of the log-likelihood is easiest here:

$$\frac{\partial^2}{\partial(\sigma^2)^2} l(\sigma^2) = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6}\sum(X_i - \mu)^2$$

and its mean equals

$$\mathbf{E}\left(\frac{\partial^2}{\partial(\sigma^2)^2} l(\sigma^2)\right) = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \cdot n\sigma^2 = -\frac{n}{2\sigma^4}.$$

We conclude that the CRLB for a smooth unbiased estimator of σ^2 is $\frac{2\sigma^4}{n}$.

Remark: We know that

$$S^2 = \frac{1}{n-1}\sum (X_i - \overline{X})^2$$

is unbiased for σ^2. What is $\mathbf{Var}\,(S^2)$? Does S^2 attain the CRLB?

We may use the fact that $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. Then

$$\mathbf{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1),$$

so

$$\mathbf{Var}\,(S^2) = \left(\frac{\sigma^2}{n-1}\right)^2 \cdot 2(n-1) = \frac{2\sigma^4}{n-1} > \text{ CRLB}$$

and S^2 does not attain the CRLB.

Discrete random variables The CRLB also holds for discrete random variables under some regularity assumptions on the probability mass function (these are technical conditions related to uniform convergence of derivatives w.r.t. θ).

In other words, let $\underline{X}$ be a discrete random sample from a distribution with probability mass function $f_X(x;\theta)$. Let $T(\underline{X})$ be an unbiased estimator of θ and assume the required regularity assumptions. Then

$$\mathbf{V}ar\,(T(\underline{X})) \geq \frac{1}{I(\theta)} = \frac{1}{\mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_{\underline{X}}(\underline{X};\theta)\right]^2\right)}\,.$$

The bound can also be expressed as

$$\mathbf{V}ar\,(T(\underline{X})) \geq \frac{-1}{\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_{\underline{X}}(\underline{X};\theta)\right)}\,.$$

More challenging material: Proof of the CRLB

We are ready to derive the Cramer–Rao Lower Bound for unbiased estimators of θ. Let $\rho(T,U)$ be the correlation between two random variables T and U:

$$\rho(T,U) = \frac{\mathbf{Cov}\,(T,U)}{\sqrt{\mathbf{Var}\,(T)\mathbf{Var}\,(U)}}\,. \tag{2}$$

The proof of the CRLB exploits the fact that $-1 \leq \rho(T,U) \leq 1$ for any T and U. We therefore have, by squaring (2),

$$\rho(T,U)^2 = \frac{\mathbf{Cov}\,(T,U)^2}{\mathbf{Var}\,(T)\mathbf{Var}\,(U)} \leq 1\,.$$

Equivalently, by rearranging,

$$\mathbf{Var}\,(T) \geq \frac{\mathbf{Cov}\,(T,U)^2}{\mathbf{Var}\,(U)}\,. \tag{3}$$

We will let T be $T(\underline{X})$, the unbiased estimator of θ in the statement of the CRLB. We will take U to be the score function

$$U(\theta, \underline{X}) = \frac{\partial}{\partial\theta} \log f_{\underline{X}}(\underline{X}; \theta)\,.$$

We will show that

(i) $\mathbf{E}(U) = 0$, and

(ii) $\mathbf{Cov}\,(T, U) = 1$.

These results will allow us to prove the CRLB.

(i) Beginning with $\mathbf{E}(U)$:

$$\begin{aligned}
\mathbf{E}(U) &= \int \left(\frac{\partial}{\partial\theta} \log f_{\underline{X}}(\underline{x}; \theta)\right) f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} \\
&= \int \left(\frac{1}{f_{\underline{X}}(\underline{x}; \theta)}\right) \left(\frac{\partial}{\partial\theta} f_{\underline{X}}(\underline{x}; \theta)\right) f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} \\
&= \int \frac{\partial}{\partial\theta} f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x}\,.
\end{aligned}$$

Now, we assume A1 and apply the Leibniz integral rule to interchange the derivative (with respect to θ) and the integral (with respect to x), i.e.,

$$\int \frac{\partial}{\partial\theta} f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} = \frac{\partial}{\partial\theta} \int f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x}\,. \tag{4}$$

Then, since

$$\frac{\partial}{\partial\theta} \int f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} = \frac{\partial}{\partial\theta} 1 = 0\,,$$

$\mathbf{E}(U) = 0$, as required.

Note that equation (4) does not hold in all cases; for example, it might not hold if the support of the distribution of X depends on θ.

(ii) Note that

$$\int T(\underline{x}) f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} = \theta\,,$$

(since $T(\underline{X})$ is unbiased). Differentiating with respect to θ and assuming we can swap the derivative and integral (this is the regularity condition mentioned in the statement) we have

$$\int T(\underline{x}) \frac{\partial}{\partial\theta} f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} = 1 \Rightarrow \int T(\underline{x}) \left(\frac{\partial}{\partial\theta} \log f_{\underline{X}}(\underline{x}; \theta)\right) f_{\underline{X}}(\underline{x}; \theta)\, d\underline{x} = 1\,,$$

so that $\mathbf{E}(TU) = 1$.

Hence,

$$\mathbf{Cov}\,(T, U) = \mathbf{E}(TU) - \mathbf{E}(T)\mathbf{E}(U) = 1\,.$$

Using (i) and (ii) above, our lower bound (3) becomes

$$\mathbf{Var}\,(T) \geq \frac{1}{\mathbf{Var}\,(U)} = \frac{1}{\mathbf{E}(U^2)}\,.$$

Note that when the X_i are i.i.d.

$$\begin{aligned} U(\theta, \underline{X}) &= \frac{\partial}{\partial\theta} \log f_{\underline{X}}(\underline{X}; \theta) \\ &= \frac{\partial}{\partial\theta} \sum_{i=1}^{n} \log f_X(X_i; \theta) \\ &= \sum_{i=1}^{n} U_i(\theta, X_i)\,. \end{aligned}$$

Since the X_i are i.i.d., so are the U_i.

Now, we have that $\mathbf{E}(U_i) = 0$ for all i, so

$$\begin{aligned} \mathbf{Var}\,(U(\theta, \underline{X})) &= \sum_{i=1}^{n} \mathbf{Var}\,(U_i(\theta, X_i)) \\ &= n\mathbf{Var}\,(U_1(\theta, X_i)) \\ &= n\mathbf{E}([U_1(\theta, X_i)]^2) \\ &= n\mathbf{E}\left(\left[\frac{\partial}{\partial\theta} \log f_{X_1}(X; \theta)\right]^2\right)\,. \end{aligned}$$

Lastly, as mentioned previously, discrete random variables require a different set of regularity conditions, which are technical and not covered in the course.

Exercise: Extension to biased estimators

Modify the proof of the CRLB for unbiased estimators presented above to show that if $T(\underline{X})$ is biased, then the followed CRLB holds:

$$\mathbf{Var}\,(T(\underline{X})) \geq \frac{\left(\frac{\partial}{\partial\theta}\mathbf{E}\,(T(\underline{X}))\right)^2}{n\mathbf{E}\left(\left[\frac{\partial}{\partial\theta} \log f_X(X; \theta)\right]^2\right)}\,.$$

Note that the numerator in this expression may be written as

$$\left(\frac{\partial}{\partial\theta}\mathbf{E}\,(T(\underline{X}))\right)^2 = \left(1 + \frac{\partial}{\partial\theta}B(T)\right)^2\,,$$

where $B(T)$ is the bias of T, since $\mathbf{E}(T(\underline{X})) = \theta + B(T)$.

As a follow-up question, under what conditions on $B(T)$ would T potentially outperform unbiased estimators of θ in terms of estimation variance?

Minimum Variance Unbiased Estimators

Minimum variance unbiased estimators are unbiased estimators that have the smallest possible variance for the considered problem. If an unbiased estimator attains the CRLB then it must be the MVUE, as there can't be an unbiased estimator with smaller variance. The converse is not true, as there are problems for which no estimator attains the CRLB (in fact, some problems do not even admit unbiased estimators!). We will restrict our discussion to estimators that are continuously differentiable - extensions to non-smooth estimators are possible but highly technical and beyond of the scope of the course.

What are the necessary and sufficient condition for a smooth unbiased estimator to attain the CRLB?

CRLB Corollary - Minimum Variance Unbiased Estimators: Let $\underline{X}$ be a random sample from a distribution satisfying the regularity conditions of the CRLB. Let $T(\underline{X})$ be a smooth unbiased estimator of θ. Then $T(\underline{X})$ attains the CRLB if and only if $U(\theta, \underline{X})$ can be expressed in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]$$

for some function $a(\theta)$.

Moreover, if $U(\theta, \underline{X})$ can be written in the form

$$U = a(\theta)\,[T - \theta]\ ,$$

where $T = T(\underline{X})$ is any statistic, then

(i) T is unbiased for θ.

(ii) T attains the CRLB.

(iii) $a(\theta) = I(\theta)$, the information, so that $\mathbf{Var}\,(T) = 1/a(\theta)$.

Note that since the CRLB corollary is an 'if and only if' result, it implies that $T(\underline{X})$ must be the MLE $\widehat{\theta}$, since this solves $U = 0$, i.e., $l'(\theta) = 0$.

Therefore, we have the useful corollary that the MLE $\widehat{\theta}$ is the MVUE if and only if $U(\theta, \underline{X})$ can be expressed in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]\ .$$

Similarly, there can only be one way of writing the score function in the form

$$U(\theta, \underline{X}) = a(\theta)\,[T(\underline{X}) - \theta]\ ,$$

so $T = T(\underline{X})$ must be the *unique* MVUE if such a representation of the score function exists.

Lastly, regarding changes in the parametrisation of the statistical model, suppose an alternative parametrisation $\phi = \phi(\theta)$. Then the existence of a MVUE for θ that attains the CRLB *does not imply* the existence of a similar MVUE for ϕ, unless $\phi(\theta)$ is a linear function.

Example 3.1 revisited

In this example, $\underline{X}$ was a random sample from the normal distribution $\mathrm{N}(\mu, \sigma^2)$, where σ^2 is known and we wish to estimate μ. We found the score function

$$U(\mu, \underline{X}) = \frac{n}{\sigma^2}\left(\overline{X} - \mu\right) ,$$

so that $\widehat{\mu} = \overline{X}$ is the MLE, is unbiased, and attains the CRLB with variance $\mathbf{Var}\left(\widehat{\mu}\right) = \sigma^2/n$.

Example 3.2 revisited

In this example, $\underline{X}$ was a random sample from the normal distribution $\mathrm{N}(\mu, \sigma^2)$, where μ is known and we wish to estimate σ^2. We found the score function

$$\begin{aligned} U(\sigma^2, \underline{X}) &= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(X_i - \mu)^2 \\ &= \frac{n}{2\sigma^4}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - \sigma^2\right] , \end{aligned}$$

so that $\widehat{\sigma^2} = \frac{1}{n}\sum(X_i - \mu)^2$ is unbiased, and attains the CRLB with variance $\mathbf{Var}\left(\widehat{\sigma^2}\right) = 2\sigma^4/n$.

Note that this requires the parameter μ to be known. If both μ and σ^2 are unknown (which is common in practical situations), then the MLE of σ^2 is

$$\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 ,$$

which is biased. In this case

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

is an unbiased estimator of σ^2, but as we have seen previously, this does not attain the CRLB.

More challenging material: Proof of the CRLB Corollary related to MVUEs

The inequality in the CRLB came from the fact that $\rho(T, U)^2 \leq 1$. We therefore get equality if and only if $\rho(T, U) = \pm 1$, in which case T and U are linearly related:

$$U = a(\theta)T + b(\theta) .$$

Since $\mathbf{E}(T) = \theta$ and $\mathbf{E}(U) = 0$, taking expectations gives

$$0 = a(\theta)\theta + b(\theta) \qquad \Rightarrow \qquad b(\theta) = -a(\theta)\theta ,$$

and therefore $U = a(\theta)[T - \theta]$, as required. □

Moreover, regarding the extension that applies for *any* statistic T:

(i) This is immediate from the fact that $\mathbf{E}(U) = 0$.

(ii) This follows from the Lemma above.

(iii) To see this, differentiate $U(\theta, \underline{X})$:

$$\begin{aligned} \frac{\partial}{\partial\theta} U(\theta, \underline{X}) &= \frac{\partial}{\partial\theta} \left(a(\theta) \left[T - \theta \right] \right) \\ &= a'(\theta) \left[T - \theta \right] - a(\theta) \,. \end{aligned}$$

Now, $\mathbf{E}(T) = \theta$, and so

$$I(\theta) = -\mathbf{E} \left(\frac{\partial}{\partial\theta} U(\theta, \underline{X}) \right) = a(\theta)$$

Asymptotic properties of Maximum Likelihood Estimators

There is a rich literature on the theoretical properties of maximum likelihood estimation (MLE), particularly for models that verify the regularity conditions that we required to prove the CRLB. In this unit we highlight the following useful results:

Consistency of the MLE: Let $\underline{X}_n$ be an i.i.d. random sample of size n from a distribution with density function $f_X(x;\theta)$ verifying appropriate regularity conditions (similar to the conditions required for the CRLB to hold), and let $\widehat{\theta}(\underline{X}_n)$ be the MLE computed from $\underline{X}_n$. Then, $\widehat{\theta}(\underline{X}_n)$ is a consistent estimator of θ.

Asymptotic normality of the MLE: Let $\underline{X}_n$ be an i.i.d. random sample of size n from a distribution with density function $f_X(x;\theta)$ verifying appropriate regularity conditions, and let $\widehat{\theta}(\underline{X}_n)$ be the MLE computed from $\underline{X}_n$. Then,

$$\frac{\widehat{\theta}(\underline{X}_n) - \theta}{\sqrt{I(\theta)^{-1}}} \to \mathrm{N}(0,1) \,, \quad \text{as } n \to \infty \,.$$

This result is a very powerful, as it says that, **asymptotically**, MLEs:

- are unbiased,
- are normally distributed, and
- attain the CRLB.

Moreover, we often apply this result as an approximation when the sample size n is large. That is, we assume that

$$\frac{\widehat{\theta}(\underline{X}_n) - \theta}{\sqrt{I(\theta)^{-1}}} \sim \mathrm{N}(0,1) \,,$$

even if this is not generally true for finite n.

For example, we use this approximation to obtain a pivotal quantity for θ to construct an approximate 95% confidence interval for θ from the MLE:

$$\widehat{\theta} \pm 1.96\sqrt{I(\widehat{\theta})^{-1}}\,.$$

where, in addition to the normal distribution approximation, we have also approximated $I(\theta)$ by $I(\widehat{\theta})$ in the denominator of the pivot. The error introduced by such approximations can be straightforwardly assessed by using simulation in R.

Example 3.3

Let $\underline{X}$ be a random sample from an Exp(λ) distribution with pdf $f_X(x;\lambda) = \lambda e^{-\lambda x}$ for $x > 0$.

We will use the MLE $\widehat{\lambda}$ and the asymptotic normality of the MLE to obtain an approximate large-sample 95% CI for λ.

Recall the MLE $\widehat{\lambda} = 1/\overline{X}$. You can check that for this model $I(\lambda) = \frac{n}{\lambda^2}$. Therefore,

$$\frac{\widehat{\lambda} - \lambda}{\sqrt{\lambda^2/n}} \approx N(0,1)$$

for large n. Using this as a pivot, we get

$$\mathbf{P}\left(-1.96 < \sqrt{n}\left(\frac{\widehat{\lambda}}{\lambda} - 1\right) < 1.96\right) = 0.95,$$

and the rearrangement leads to

$$\frac{\widehat{\lambda}}{1 + \frac{1.96}{\sqrt{n}}} < \lambda < \frac{\widehat{\lambda}}{1 - \frac{1.96}{\sqrt{n}}}$$

as the required (approximate) 95% CI.

Data: Suppose $n = 10$ and $\overline{x} = 1.73$. Then $\widehat{\lambda} = 1/1.73 = 0.578$ and the CI is

$$\left(\frac{0.578}{1 + \frac{1.96}{\sqrt{10}}}, \frac{0.578}{1 - \frac{1.96}{\sqrt{10}}}\right) = (0.357, 1.520)$$

(which is not symmetric about $\widehat{\lambda}$).

An alternative is to use a "double approximation" where we replace $I(\lambda) = \frac{n}{\lambda^2}$ by $I(\widehat{\lambda})$ in the construction of the confidence interval. More precisely, we assume that the pivot

$$\frac{\widehat{\lambda} - \lambda}{\sqrt{\widehat{\lambda}^2/n}} \sim N(0,1)$$

(this is an approximation that only holds exactly as n increases to infinity). Then the approximate 95% CI for λ is

$$\widehat{\lambda} \pm 1.96\frac{\widehat{\lambda}}{\sqrt{n}}.$$

In particular, for our data, the CI is

$$0.578 \pm 0.358 = (0.220, 0.936)\,,$$

which is quite different!

Note: $n = 10$ is **not large**, so these CI's must be considered as "very approximate".

Suppose instead that $n = 100$ and $\overline{x} = 1.73$. Then the first CI is $(0.483, 0.719)$, and the second CI is $(0.465, 0.691)$. They are similar now!

Asymptotic properties of the deviance function

Let $\underline{X}_n$ be an i.i.d. random sample of size n from a distribution with density function $f_X(x;\theta)$ verifying regularity conditions similar to the ones required for the CLRB to hold. Let $\widehat{\theta}(\underline{X}_n)$ denote the MLE of θ computed from $\underline{X}_n$. Our final result in this unit derives the asymptotic distribution for an important function, called the *deviance*, given for any θ by

$$D(\theta) = 2\left[l(\widehat{\theta}(\underline{X}_n)) - l(\theta)\right]\,.$$

In other words, this function measures how close a value θ is from the MLE $\widehat{\theta}$ by comparing the likelihoods at θ and $\widehat{\theta}$.

Note that the deviance as stated above is a random variable because it is a function of the random variable $\underline{X}_n$. The following results describe the distribution of this random variable as n increases to infinity.

Asymptotic properties of the deviance: As $n \to \infty$, the deviance $D(\theta)$ converges to a χ^2_1 random variable (i.e., a chi-squared random variable with one degree of freedom).

Corollary: Suppose $\chi^2_1(\alpha)$ is such that

$$\mathbf{P}(\chi^2_1 > \chi^2_1(\alpha)) = \alpha\,.$$

Then an approximate large-sample $100(1-\alpha)\%$ confidence interval for θ is

$$\left\{\theta : D(\theta) < \chi^2_1(\alpha)\right\}\,.$$

Now, $\chi^2_1(0.05) = 3.841 (= 1.96^2)$, and so an approximate 95% confidence interval can be obtained as the set of those θ for which $D(\theta) < 3.841$, or, equivalently, those θ for which $l(\theta)$ lies within 1.92 of the maximal value $l(\widehat{\theta})$.

Example 3.3 revisited

In the exponential setting of Example 3.3, we had the data $n = 10$ and $\sum x_i = 17.3$. We now use the above corollary to calculate an approximate 95% confidence interval for λ.

Recall that

$$l(\lambda) = n \log \lambda - \lambda \sum_{i=1}^{n} x_i \quad \text{and} \quad D(\lambda) = 2(l(\hat{\lambda}) - l(\lambda)).$$

For the 95% confidence interval, λ should be such that $D(\lambda) < 3.841$. See the R code below, which gives the 95% confidence interval $(0.29, 1.01)$.

Note that this confidence interval is in between the two obtained earlier.

For the $n = 100$ case, the corresponding 95% CI is $(0.473, 0.698)$ – again in between the two others (this is an exercise for you!).

```
n = 10 #set sample size
xbar = 1.73 #observed sample mean

thresh = qchisq(0.95,1) #threshold for deviance to define CI
lam = 0.1+c(1:1000)*0.001 #set up vector of lambda values
dev = 2*(lam*n*xbar-n*(log(xbar)+log(lam)+1)) #calculate deviance
devci = dev[dev<thresh] #consider values below thresh
lamci = lam[dev<thresh] #and corresponding lambda values

lamci[1] #left end point of CI
lamci[length(devci)] #right end point of CI
```

This R code produces the following output:

```
> lamci[1]
[1] 0.29
> lamci[length(devci)]
[1] 1.013
```

Appendix

We seek to establish that under A2,

$$\mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_X(X;\theta)\right]^2\right) = -\mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_X(X;\theta)\right).$$

We know that the score function $U(\theta, X) = \frac{\partial}{\partial\theta}\log f_X(X;\theta)$ has expected value zero. That is,

$$0 = \int f_X(x;\theta)\frac{\partial}{\partial\theta}\log f_X(x;\theta)\,dx\,.$$

Differentiating both sides of this equation, we get

$$0 = \frac{\partial}{\partial\theta}\left(\int f_X(x;\theta)\frac{\partial}{\partial\theta}\log f_X(x;\theta)\,dx\right).$$

We now use A2 to apply the Leibniz integral rule to interchange the derivative (with respect to θ) and the integral (with respect to x). Then, using the product rule to differentiate the function inside the integral above, we get

$$\begin{aligned}
0 &= \int \frac{\partial}{\partial\theta}f_X(x;\theta)\frac{\partial}{\partial\theta}\log f_X(x;\theta)\,dx + \int f_X(x;\theta)\frac{\partial^2}{\partial\theta^2}\log f_X(x;\theta)\,dx \\
&= \int f_X(x;\theta)\frac{\partial}{\partial\theta}\log f_X(x;\theta)\frac{\partial}{\partial\theta}\log f_X(x;\theta)\,dx + \int f_X(x;\theta)\frac{\partial^2}{\partial\theta^2}\log f_X(x;\theta)\,dx \\
&= \int f_X(x;\theta)\left(\frac{\partial}{\partial\theta}\log f_X(x;\theta)\right)^2 dx + \int f_X(x;\theta)\frac{\partial^2}{\partial\theta^2}\log f_X(x;\theta)\,dx \\
&= \mathbf{E}\left(\left[\frac{\partial}{\partial\theta}\log f_X(X;\theta)\right]^2\right) + \mathbf{E}\left(\frac{\partial^2}{\partial\theta^2}\log f_X(X;\theta)\right).
\end{aligned}$$